

CS 4412 M3 Complete Implementation

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I. DATASET DESCRIPTION

My main datasets are the IMDb top 250 movies and Box Office Mojo's top 200 highest grossing movies, with Rotten Tomatoes acting as a bridge between them. Each of my sources represents the main points of my discovery questions; IMDb has the general response, BOM has the profits, and RT has the individual audience and critic responses. An important thing to note about the datasets is that not every movie is shared between them, as in some films were hated but profitable while others became forgotten classics. Instead of fixing this gap, I plan to embrace it. After all, blockbusters and critical darlings are the lifeblood of the industry.

The end result is a catalog of around 450 movies of various responses and profits. Each movie will have 5 columns. The main three are their box office, their critical reception, and their audience receptions. These categories and the ways they intersect are what I am analyzing for my project. I have included titles and year of release just for the sake of formatting.

Frozen II	\$1,453,683,476	77%	92%	2019
Léon: The Professional	\$20,330,788	76%	95%	1994
The Chronicles of Narnia: The Lion, the Witch and the Wardrobe	\$745,013,115	76%	61%	2005
Harry Potter and the Deathly Hallows: Part 1	\$960,858,478	76%	85%	2010
Avatar: The Way of Water	\$2,334,484,620	76%	92%	2022
Lock, Stock and Two Smoking Barrels	\$3,753,929	75%	93%	1998
The Intouchables	\$426,590,315	75%	93%	2011
Forrest Gump	\$678,226,465	75%	95%	1994
Forrest Gump	\$678,226,465	75%	95%	1994

Fig. 1. Raw Dataset. Note the duplicate at the bottom.

II. DISCOVERY QUESTIONS

The big questions driving my project are:

1. "What kind of films emerge based on the critic and audience opinion?"
2. "What mixture of ratings and revenue distinguishes a 'cult

classic' from a 'blockbuster'?"

3. "How do good reviews keep a movie alive in the public consensus?"

I wanted to ask these questions because of my extended contact with the media discussion side of the internet, for better or worse. I have seen people claim that a highly profitable film with good critical reviews had bribed reviewers because it had low audience ratings, and I have seen those same people claim that a profitable film with low critical ratings has a cult audience blindly giving the media their money. This resulted in the first two questions being asked to find a solid answer. Besides that, I also wanted to discuss There is a term known as "Vindicated by History", where a work that failed in the former present becomes heavily praised in the current present. I was reminded of this phrase after I looked through my datasets and found that a sizable number of famous movies do not make that much of a profit. So, I decided to establish another study on the differences between financial impact and cultural impact.

With the implementation finished, I can finally answer these questions based on what I have found.

A. Question 1: Movie Patterns

The four main patterns of movies are Excellent, Biased, Mixed, and Panned. Excellent Movies are given high praise from both group, regardless of how many members went out to see them in theaters. Biased Movies are given good treatment from both groups but one side enjoys it more. Mixed Movies have good ratings from one side and middling reviews from the other. Panned Movies are negatively received from both types of film-goers.

B. Question 2: Profit Patterns

The line between a smash hit and a stealth success is small in length but large in width. The length of the line takes the form of positive reception, which can be gotten from every critic enjoying an unpopular film or from most critics liking a famous film. Meanwhile, the width of the divider takes the form of negative reception. Since opinions only form after watching the movies, bad opinions feed the box office. Cult classics that are enjoyed by the few who see them and blockbusters that attract millions and print millions more are separated by movies that are liked by enough groups of people to make a profit.

C. Question 3: Longevity

Good reviews can keep older movies alive in the mind of the public much more than the box office can. This can manifest in

multiple ways. Sometimes it comes in the form of franchises or inspired media, but that has the major risk of producing something that audiences don't enjoy. Other times, it is through releases. This is why there is a single spike in profitable movies around 1939. The most important way reviews fuel old movies is the way invoked in the previous question- cult classics that people enjoy watching and talk about it to others so that they can watch it to.

III. EXPLORATORY DATA ANALYSIS

Originally, I wanted to create a punnet square to sort the movies according to their audience and reviewer scores. This would have reflected the Rotten Tomatoes scoring system but not simulate it, as I planned for it to follow my k-means results. However, after going through EDA, I have chosen to follow some more solid graphs.

Once all of the films have been recorded and defined, the real analysis begins. I have a few plots for each column that help establish some findings. The box-plots show that most movies earn under a billion dollars, usually half that, with the major money makers being extreme outliers. Meanwhile, critics and audiences tend to enjoy what they watch but have different standards for what they tend to enjoy.

Meanwhile, the bar graphs show just how big the gap between profit and popularity really is. The number of low-profit movies compared to everything else, even the ones in the hundreds of millions, combined with the peak reviews being in the good to great range implies the preference of classics over blockbusters. However, my data could have been heavily twisted by outside factors such as the demographic of moviegoers across the years or people's personal biases affecting the total reception.

A. Data Preprocessing

Due to taking data from two separate sites that discuss the profits and reception of movies respectively, there were a few duplicates. An interesting thing to note is that most of the duplicates had some kind of brand attachment, like Pixar for *Finding Nemo* or Tom Hanks for *Forrest Gump*. The *Fast and Furious* franchise lacking any repeats while the *Jurassic Park* series only has the first film duplicated implies a required amount of skill and marketing to appear on both lists.

As for true outliers, the only major one is a 1975 Turkish comedy named *The Chaos Class Failed the Class*, which appears to be a complete anomaly. The minor outliers has a missing box office number or critical reception, but these can be fixed through third party sources.

B. Overlap

Looking at where the data point clump together forms a really intriguing picture. When comparing profits and critical reception, there appears to be a universal standard profit or 800 million dollar that is formed across all metrics of reviews. In the box graphs for year of release, films has slowly risen their profit limits so that what was the maximum in the 80s or 90s is the standard of the 2010s. Meanwhile, the increase

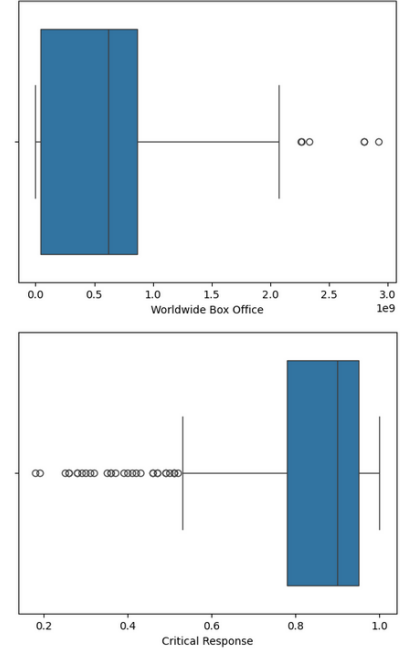


Fig. 2. Profit and Critical Response Box Plots

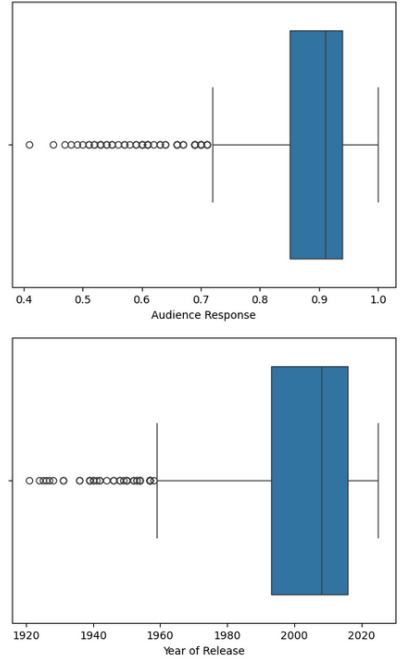


Fig. 3. General Response and Release Year Box Plots

in film releases means that more underwhelming films are being watched as well. Finally, and once again, audiences and critics can agree on what is excellent but disagree on what is mediocre.

IV. TECHNIQUES

I started by using clustering to mine through my datasets to help smooth down the rough data into my precise rankings. I built a k-means algorithm to find the optimal number of clusters. My elbow-graph result found three to be the best match, so I plugged that amount of clusters into the model

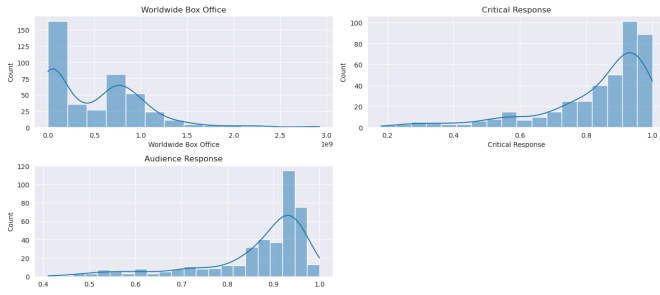


Fig. 4. Average Distribution of Box Office (Top Left), Critical Response (Top Right), and Audience Response (Bottom)

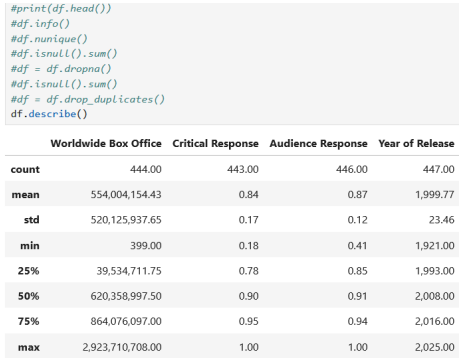


Fig. 5. Data before Preprocessing

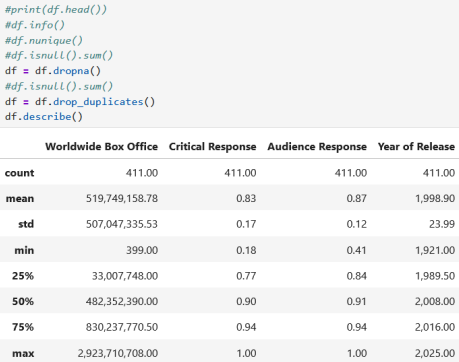


Fig. 6. Data after Preprocessing

and ran it ten times. Two of the clusters were overlapping each other, which I ignored until I tried to find the centroids. After cross-referencing my two kmeans functions with my elbow graph, I realized that there might have been 4 clusters in my data.

For M3, I expanded into decision trees to further strengthen my data. The idea was to split the data by the box office and see what patterns of reception can be formed. For example, I could split the data by less than 1 billion dollars in profit and calculate the gain achieved from that. Unfortunately, the only class which could function in the decision tree were the clusters, which was against the rules. As such, I aim to add another variable to my data set just for this tree.

V. RESULTS

The first cluster, dubbed 'Minivan', consists of movies that are usually seen by groups of people. These films are meant

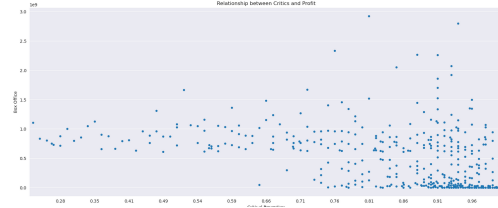


Fig. 7. Swarm Plot of Critics versus Profits



Fig. 8. Box Graph of Year of Release versus Critical Response



Fig. 9. Box Graph of Year of General versus Professional Responses

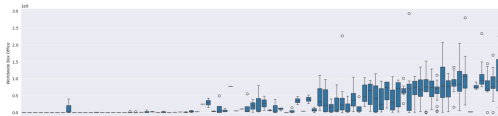


Fig. 10. Box Graph of Year of Release versus Box Office

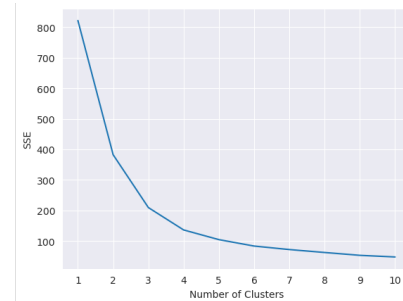


Fig. 11. K-Means Elbow Graph

for multiple different audiences, like families or people who want to see it as a joke. They also prefer profits over reactions. The second cluster, dubbed 'Classic', holds movies that everyone remembers due to pop culture osmosis. The age of the data might be why the box office is so low since inflation does not apply to them. With these three clusters formed, I can build a more detailed map later on that can analyze a film's impact and legacy.

The third cluster, dubbed 'Blockbuster', has the films that brought in tons of profit despite some push-back from either audiences or critics. Each element have a box office of over 1.3 billion dollars, they usually have an association with an wide-reaching franchise, and all but one were released in the 21st century.

The final cluster, dubbed 'Genre', is populated by well-received movies with very specialized aesthetics. The more profitable end of the cluster is filled with famous brands like Star Wars and Pixar that bring more people in than their

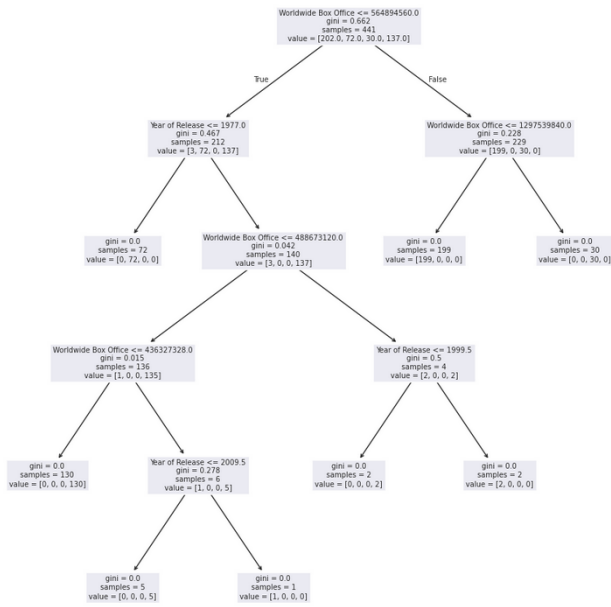


Fig. 12. Draft of Decision Tree

	Film Name	Worldwide Box Office	
24	Avatar	2,923,718,708.00	
25	Avatar: Fire and Ash	1,488,738,910.00	
26	Avatar: The Way of Water	2,334,484,620.00	
27	Avengers: Age of Ultron	1,485,818,048.00	
28	Avengers: Endgame	2,799,439,100.00	
29	Avengers: Endgame	2,799,439,100.00	
30	Avengers: Infinity War	2,052,415,039.00	
31	Avengers: Infinity War	2,052,415,039.00	
33	Barbie	1,447,138,421.00	
43	Black Panther	1,349,926,083.00	
71	Deadpool & Wolverine	1,328,873,645.00	
110	Frozen II	1,453,683,476.00	
113	Furious 7	1,515,942,457.00	
134	Harry Potter and the Deathly Hallows: Part 2	1,342,942,050.00	
135	Harry Potter and the Deathly Hallows: Part 2	1,342,942,050.00	
164	Inside Out 2	1,698,863,816.00	
179	Jurassic World	1,671,537,444.00	
181	Jurassic World: Fallen Kingdom	1,388,566,455.00	
229	Ne Zha 2	2,259,822,417.00	
292	Spider-Man: No Way Home	1,921,426,073.00	
293	Spider-Man: No Way Home	1,921,426,073.00	
305	Star Wars: Episode VII - The Force Awakens	2,071,310,218.00	
306	Star Wars: Episode VIII - The Last Jedi	1,334,407,766.00	
315	The Avengers	1,528,538,536.00	
363	The Lion King (2019)	1,662,828,819.00	
388	The Super Mario Bros. Movie	1,368,979,735.00	
410	Titanic	2,264,812,968.00	
414	Top Gun: Maverick	1,495,696,292.00	
415	Top Gun: Maverick	1,495,696,292.00	
446	Zootopia 2	1,859,358,209.00	

	Critical Response	Audience Response	Year of Release	cluster
24	0.81	0.82	2009	2
25	0.66	0.90	2025	2
26	0.76	0.92	2022	2
27	0.75	0.82	2015	2
28	0.94	0.90	2019	2
29	0.94	0.90	2019	2
30	0.85	0.92	2018	2
31	0.85	0.92	2018	2
33	0.88	0.83	2023	2
43	0.96	0.79	2018	2
71	0.78	0.94	2024	2
110	0.77	0.92	2019	2
113	0.81	0.82	2015	2
134	0.96	0.89	2011	2
135	0.96	0.89	2011	2
164	0.91	0.94	2024	2
179	0.72	0.78	2015	2
181	0.47	0.48	2018	2
229	0.91	0.98	2025	2
292	0.93	0.97	2025	2
293	0.93	0.97	2025	2
305	0.93	0.84	2015	2
306	0.91	0.41	2017	2
315	0.91	0.91	2012	2
363	0.52	0.88	2019	2
388	0.59	0.95	2023	2
410	0.88	0.69	1997	2
414	0.96	0.99	2022	2
415	0.96	0.99	2022	2
446	0.91	0.96	2025	2

Fig. 14. The Blockbuster Cluster

	Film Name	Worldwide Box Office	Critical Response	
0	1917	446,864,352.00	0.88	
1	2012	791,217,826.00	0.40	
9	A Minecraft Movie	961,187,780.00	0.47	
12	Aladdin (2019)	1,054,304,000.00	0.57	
13	Alice in Wonderland (2010)	1,025,468,216.00	0.51	
..	...	...	...	
439	Wolf Warrior 2	870,325,439.00	0.67	
440	Wonder Woman	823,978,682.00	0.93	
441	Wonka	634,582,312.00	0.82	
442	X-Men: Days of Future Past	746,045,700.00	0.90	
445	Zootopia	1,025,521,689.00	0.98	

	Audience Response	Year of Release	cluster
0	0.88	2019	0
1	0.47	2009	0
9	0.84	2025	0
12	0.94	2019	0
13	0.55	2010	0
..	...	...	...
439	0.74	2017	0
440	0.83	2017	0
441	0.90	2023	0
442	0.91	2014	0
445	0.92	2016	0

[202 rows x 6 columns]

	Film Name	Worldwide Box Office	Critical Response	
2	12 Angry Men	9,844.00	1.00	
5	2001: A Space Odyssey	68,010,425.00	0.90	
8	A Clockwork Orange	27,166,998.00	0.86	
16	All About Eve	156,988.00	0.99	
34	Barry Lyndon	579,213.00	0.78	
..	...	...	...	
413	Tokyo Story	93,881.00	1.00	
430	Vertigo	7,982,466.00	0.93	
436	Wild Strawberries	184,614.00	0.94	
438	Witness for the Prosecution	7,903.00	1.00	
443	Yojimbo	68,196.00	0.96	

	Audience Response	Year of Release	cluster
2	0.97	1957	1
5	0.88	1968	1
8	0.93	1971	1
16	0.94	1950	1
34	0.92	1975	1
..	...	...	...
413	0.93	1953	1
430	0.92	1958	1
436	0.94	1957	1
438	0.95	1957	1
443	0.96	1961	1

[72 rows x 6 columns]

Fig. 13. The Classic and Minivan clusters

	Film Name	Worldwide Box Office	Critical Response	
3	12 Years a Slave	187,734,091.00	0.95	
4	12th Fail	138,288.00	0.91	
6	3 Idiots	60,262,836.00	1.00	
7	A Beautiful Mind	316,791,257.00	0.74	
10	A Separation	22,926,076.00	0.99	
..	...	...	...	
428	V for Vendetta	134,686,457.00	0.73	
432	Warrior	23,308,615.00	0.84	
434	Whiplash	50,431,161.00	0.94	
437	Wild Tales	31,478,893.00	0.94	
444	Your Name	485,320,132.00	0.98	

	Audience Response	Year of Release	cluster
3	0.90	2013	3
4	0.97	2023	3
6	0.93	2009	3
7	0.93	2001	3
10	0.92	2011	3
..	...	...	...
428	0.90	2005	3
432	0.92	2011	3
434	0.94	2014	3
437	0.92	2014	3
444	0.94	2016	3

[137 rows x 6 columns]

Fig. 15. The Genre Cluster

genres.

A. Centroid Clusters

My testing with centroids shows an alternative way to format the data points. While 'Classic', 'Minivan', and 'Blockbuster' remain the same, I have add the final category of 'Worldbreaking' to the list. This refers to the outliers of the graph and are named after who these films are well-known

for how much they made in the eyes of many people. This

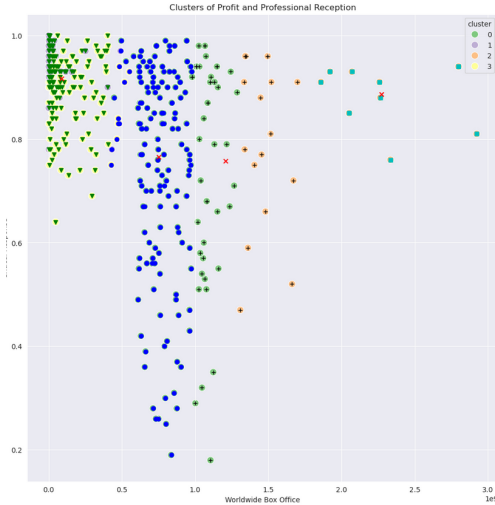


Fig. 16. The K-means centroid graph for Box Office against Critical Reception. Green Triangle for Classic, Blue Spheres for Minivan, Black Crosses for Blockbusters, and Cyan Squares for Worldbreaking

idea is better shown when comparing reception and everything gets screwy. In this scenario, the centroids represent how the two audiences react to movies; high praise, moderate praise, diverse viewpoints, and mutual dislike.

VI. M4 PLAN

My main idea for M4 is to perform the same analysis process that I did during M3 for each individual cluster. This won't extend to creating a k-means graph within a k-means graph, rather it is seeing if there are any outliers in the clusters and if any patterns from the base dataset repeat.

While we are discussing my data, I also want to polish up my work as a whole. Beside anything I may have overlooked in this installment, there might be another way to expand my research. Depending on how much time I have to improve, I could update my original data with some external variables. Currently I am picturing adding a list of genres for each film and cross-referencing that with my clusters to see if another pattern appears.

Finally, I want to polish up the document by having the figures in the places where I want them to be. Even if I use the terminology that should hold an image in place, Overleaf just sometimes sends the images where ever it wants. It even slices up my paragraphs which breaks up the flow. This has been bugging me ever since M1 and I don't get why it's even happening. The fourth page of this report is 90 percent graphs that mashed together so that the context is missing I hate it.

VII. SOURCES

Github Link

Griffin Vines, "GitHub - PckCalc-Kraftwrk/CS4412-Project," GitHub, 2025. <https://github.com/PckCalc-Kraftwrk/CS4412-Project> (accessed April. 02, 2026).



Fig. 17. Scatterplot of General Audiences versus Critics with clusters. Not the clump of multiple different data points in the top right.

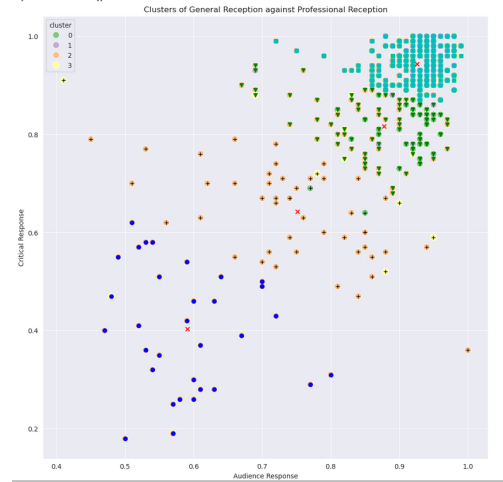


Fig. 18. Scatterplot of General Audiences versus Critics with centroids. Cyan is widely praised, green is movies where one side likes it more than the other, black is movies where one side likes it but the other side doesn't, and blue is mutual dislike

Dataset links

IMDb, "IMDb Top Rated Movies," IMDb, 2019. <https://www.imdb.com/chart/top/>

[2] "Top Lifetime Grosses - Box Office Mojo," Box Office Mojo, 2019. https://www.boxofficemojo.com/chart/ww_top_lifetime_gross/?area=XWW

[3] Rotten Tomatoes, "Rotten Tomatoes: Movies — TV Shows — Movie Trailers — Reviews," Rotten Tomatoes, 2025. <https://www.rottentomatoes.com/>

[4] GeeksforGeeks, "Data Preprocessing in Python," GeeksforGeeks, Oct. 29, 2017. <https://www.geeksforgeeks.org/data-analysis/data-preprocessing-machine-learning-python/>

[5] Zach, "K-Means Clustering in Python: Step-by-Step Example," Statology, Aug. 31, 2022. <https://www.statology.org/k-means-clustering-in-python/>

[6] J. Chen, "Seaborn Swarmplot Axis Interval Formatting," Stack Overflow, Jul. 31, 2020. <https://stackoverflow.com/questions/63196379/seaborn-swarmplot-axis-interval-formatting>

